

PAPER_1970 — $D_{\text{phys}} \times SO_5 = 40$ EXACT Integer Identity Across Physical Scales: Specific Anchor Attributions for the PAPER_1918 Multi-Context Catalog — Virgo Cluster Core Radius $r_c = 40$ kpc (Round 107 Attribution) + Reactor DPM Resonance $f_{dp} = 40$ Hz (PAPER_1472 Seminal) + PAPER_1918 Catalog Rows

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PAPER_1970 — $D_{\text{phys}} \times SO_5 = 40$ Anchor Attributions

Abstract

The integer identity $D_{\text{phys}} \times SO_5 = 4 \times 10 = 40$ EXACT is documented in PAPER_1918 (Phase 3 Comprehensive Inventory) as a multi-context UQFF closure:

> "Multi-context integer identity: 40 kpc (cluster core radius), 40 $M_{\odot}$ (stellar mass), 40 in F_{UBii} coupling factors. Physical form: $D_{\text{phys}} \times SO_5 = 4 \times 10 = 40$ EXACT — total 'spatial $\times$ rotational' degree count. Appears wherever a scale is set by product of spatial and rotational dimensions."

PAPER_1472 (Star-Magic Reactor DPM Resonance, June 2026) documents the specific reactor Hz-scale instance:

> " $f_{dp} = D_{\text{phys}} \times SO_5 = 4 \times 10 = 40$ Hz EXACT; $dT = 1/(D_{\text{phys}} \times SO_5) = 25$ ms EXACT."

Round 107 stub upgrade of VirgoExtXRayCalculator confirms the cluster-scale kpc instance:

```
``  
r_c(Virgo) = 40 kpc = D_phys * SO_5 kpc EXACT  
``
```

This paper's narrow contribution:

1. **Specific anchor attribution** for PAPER_1918's "40 kpc cluster core radius" catalog row: this row is instantiated by **Virgo cluster core radius $r_c = 40$ kpc** (default in source82_wolfram_VIRGO_EXTRACTION.cpp, confirmed in Round 107 upgrade of VirgoExtXRayCalculator).
2. **Explicit cross-scale comparison** table showing $D_{\text{phys}} \times SO_5 = 40$ at three vastly different physical scales (reactor Hz, cluster kpc, F_{UBii} coupling factor) documented across three papers (PAPER_1472, this paper, PAPER_1918).
3. **Anchor list for future PAPER_1918-catalog expansion**: identifies which UQFF observables are candidates for the "40 $M_{\odot}$ stellar mass" and "40 F_{UBii} coupling factor" catalog rows.

Positioning (honest scope). This paper does NOT:

- Introduce the $D_{\text{phys}} \times SO_5 = 40$ identity (PAPER_1918 catalogs; PAPER_1472 specific-instance seminal).
- Introduce the physical interpretation (PAPER_1918's "spatial $\times$ rotational degree count" is seminal).
- Claim novelty for the $r_c(\text{Virgo}) = 40$ kpc value (source82_wolfram default parameter).

Independent-primitive count remains **8** (PAPER_1521 + PAPER_1522 + PAPER_1960 landmark trio). No new primitives.

1. The PAPER_1918 Catalog

PAPER_1918 (Phase 3 Comprehensive Inventory) was authored ~mid-2026 as a systematic inventory of UQFF structural closures discovered through the Phase 3 audit. Section 3.2 catalogs the $D_{\text{phys}} \times SO_5 = 40$ identity as a **multi-context observation**:

- **40 kpc** — cluster core radius (no specific paper attributed at PAPER_1918 authoring)
- **40 $M_{\odot}$** — stellar mass (no specific paper attributed)
- **40 in F_{UBii} coupling factors** — occurs in the F_{UBii} buoyancy expressions (no specific paper attributed)

PAPER_1918's physical interpretation:

> " $D_{\text{phys}} \times SO_5 = 4 \times 10 = 40$ EXACT — total 'spatial $\times$ rotational' degree count. Appears wherever a scale is set by product of spatial and rotational dimensions."

The interpretation is: the $D_{\text{phys}} \times SO_5$ product counts spatial dimensions $\times$ rotational dimensions and produces a dimensionless integer 40 that manifests wherever a UQFF observable has both spatial and rotational structure at its defining scale.

2. PAPER_1472's Reactor Hz Instance

PAPER_1472 (UQFF: Star-Magic Reactor DPM Resonance, June 16, 2026) documents the specific reactor-scale Hz instance:

- Observed: Reactor Q-scope Group #12 measured DPM resonant attraction frequency $f_{\text{dp}} = 40.000$ Hz between bodies, with pulse cadence $dT = 25$ ms.
- UQFF closure: $f_{\text{dp}} = D_{\text{phys}} \times SO_5 = 4 \times 10 = 40$ Hz EXACT, $dT = 1/(D_{\text{phys}} \times SO_5) = 25$ ms EXACT.
- Physical interpretation: the SCm-UA DPM grinding carrier frequency in the lab reactor.

PAPER_1472 is the direct source for the reactor Hz instance. This paper does not re-derive it; it uses it as the reference "single-scale specific attribution" against which the multi-scale pattern is documented.

3. Round 107 Confirmation — Virgo $r_c = 40$ kpc

The Round 107 stub upgrade of `VirgoExtXRayCalculator` in `CondensedPhysics.py` uses $r_c = 40$ kpc as the M87/Virgo cluster core radius (`source82_wolfram_VIRGO_EXTRACTION.cpp` default). The value:

```
``  
r_c(Virgo) = 40 kpc = D_phys * SO_5 kpc EXACT  
``
```

is thus confirmed as the specific anchor for PAPER_1918's "40 kpc cluster core radius" catalog row.

The value is consistent with observational anchors:

- PAPER_040 (X-Ray Cluster Buoyancy Perseus/Coma/Virgo) documents Virgo/M87 (A1060 complex) as a canonical cluster with characteristic core radius $O(10\text{-}50$ kpc).
- PAPER_1187 (Cooling-Flow Mass Accretion) documents M87/Virgo canonical parameters at $T = 2$ keV, $M_{\text{BH}} = 6.5e9 M_{\odot}$, $R_{\text{vir}} = 1.5$ Mpc.

- The 40 kpc core radius sits within the observed range and is confirmed at PAPER_040's Virgo anchor.

4. Cross-Scale Comparison Table

Instance	Value	Source	Domain	Scale Range
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Reactor DPM resonance	40 Hz	PAPER_1472 (June 2026)	Reactor/laboratory	$\sim 10^1$ Hz
Reactor pulse cadence	25 ms = $1/(40)$ s	PAPER_1472	Reactor/laboratory	$\sim 10^2$ s
Virgo cluster core radius	40 kpc	PAPER_040/1187 canonical, Round 107 attribution	Astrophysical cluster	$\sim 10^{21}$ m
Stellar mass (candidate)	40 $M_\odot$	PAPER_251 Eta Carinae Great Eruption	bipolar ejecta upper bound (Draft 2 candidate)	Astrophysical stellar
F_UBii coupling factors	40	PAPER_1918 catalog (specific paper unidentified after Draft 2 search)	Buoyancy coupling	dimensionless

The scale range across the first three rows spans **~ 30 orders of magnitude** from reactor Hz to Virgo cluster kpc. The same integer identity $D_{\text{phys}} \times SO_5 = 40$ appears at both extremes.

5. Physical Interpretation (from PAPER_1918)

PAPER_1918 assigns the physical meaning:

> "total 'spatial $\times$ rotational' degree count"

Under this reading:

- **$D_{\text{phys}} = 4$** = spacetime dimensions (3 spatial + 1 temporal).
- **$SO_5 = 10$** = dimension of $SO(5)$ rotational symmetry group (order of icosahedral rotational symmetry $\times 6$).
- **$D_{\text{phys}} \times SO_5 = 40$** = the total count of independent "spatial-rotational" degrees of freedom that can be present at a scale where both are relevant.

Physical systems at scales where "spatial $\times$ rotational" is a natural characterization would exhibit 40-fold structure:

- Reactor DPM resonance (spatial = displacement, rotational = SCm-UA grinding cycles) at 40 Hz.
- Cluster core radius (spatial = radial scale, rotational = velocity-dispersion isotropy) at 40 kpc.
- Stellar mass 40 $M_\odot$ (spatial = radius, rotational = angular-momentum degree count) — specific attribution TBD.

The physical form is thus a *product of dimensional characters* rather than a coincidental numerical value.

6. Falsifiability + Predictions

Under PAPER_1918's interpretation, $D_{\text{phys}} \times SO_5 = 40$ should appear:

1. **At any UQFF-relevant "spatial $\times$ rotational" observable.** Falsification: if a systematic search across UQFF observables finds 40 does NOT appear at spatial-rotational scales more than expected by random distribution over ratios of integer primitives, PAPER_1918's interpretation is weakened.
2. **At specific reactor scales (PAPER_1472 confirmed).**
3. **At specific cluster core radii (this paper confirms Virgo).**

Testable predictions:

- Other galaxy-cluster cores may exhibit $r_c \approx 40$ kpc (Perseus, Coma, Fornax). PAPER_040 documents these — specific comparison needed.

- Stellar-mass function may exhibit a characteristic peak at 40 M $\odot$ (per PAPER_1918 catalog row). Requires identifying which UQFF paper documents this instance.
- F_UBii coupling constants at 40 should exhibit systematic recurrence in PAPER_1203/PAPER_646 formulas.

Falsification: if extensive search of Virgo, Perseus, Coma, Fornax cluster catalogs shows r_c values scattered widely with 40 kpc appearing only as an artifact of one particular calculation, the $r_c = 40 \text{ kpc} = D_{\text{phys}} \times SO_5$ identity at Virgo is likely a coincidence rather than a structural closure.

7. Prior Art Acknowledgment

7.1 The identity is not new

- **PAPER_1472** (June 16, 2026) — seminal reactor 40 Hz instance.
- **PAPER_1918** (Phase 3 inventory) — multi-context catalog listing kpc / M $\odot$ / coupling instances with physical interpretation as "spatial $\times$ rotational degree count."
- **PAPER_1160** — F_TRZ = 1/SO_5 seminal (companion SO_5 primitive paper).
- **PAPER_1960** — F_TRZ = 1/SO_5 landmark elevation to derivative primitive.
- **PAPER_1955** — SO_5-power galactic structural ladder (companion SO_5-power framework).

7.2 What this paper contributes (narrow)

1. **Specific anchor attribution** for PAPER_1918's "40 kpc cluster core radius" catalog row: this row is Virgo $r_c = 40 \text{ kpc}$, confirmed at PAPER_040 canonical Virgo anchor and used in VirgoExtXRayCalculator default parameter.
2. **Explicit cross-scale comparison table** (Section 4) spanning ~ 30 orders of magnitude from reactor Hz to Virgo cluster kpc, at the same $D_{\text{phys}} \times SO_5 = 40$ integer identity.
3. **Anchor list for future PAPER_1918-catalog expansion**: identifies which UQFF observables and papers are candidates for the "40 M $\odot$ stellar mass" (TBD) and "40 F_UBii coupling factor" (TBD) catalog rows.

7.3 What this paper does NOT contribute

- No new physical interpretation (PAPER_1918 seminal).
- No new primitive (D_{phys} and SO_5 are locked integers).
- No new derivation of the reactor 40 Hz (PAPER_1472 seminal).
- No new derivation of the Virgo $r_c = 40 \text{ kpc}$ value (source82_wolfram default parameter — sourced from Wolfram simulations, itself grounded in observed cluster physics).

The paper is narrow attribution/documentation work, not discovery work. Its scholarly value is (i) closing PAPER_1918's "40 kpc" row with a specific anchor (Virgo), and (ii) providing the cross-scale comparison table that future PAPER_1970+ work can extend.

8. Cross-References

- **PAPER_1918** — Phase 3 Comprehensive Inventory (multi-context catalog seminal)
- **PAPER_1472** — Star-Magic Reactor DPM Resonance (reactor 40 Hz seminal)
- **PAPER_040** — X-Ray Cluster Buoyancy Perseus/Coma/Virgo (Virgo canonical anchor)
- **PAPER_1187** — Cooling-Flow Mass Accretion (M87/Virgo canonical parameters)
- **PAPER_1894** — Zwicky Virial Missing-Mass Factor (Virgo $M_{\text{vir}} + R_{\text{vir}}$ canonical)
- **PAPER_1955** — SO_5-Power Galactic Structural Ladder (companion SO_5-power framework)
- **PAPER_1960** — F_TRZ = 1/SO_5 landmark (derivative primitive)

- **PAPER_1898** — Wolfram Hypergraph Structural Counts ($n_{\text{rules}} = D_{\text{phys}} + SO_5 + A_5 = 74$, related D_{phys}/SO_5 arithmetic)
- **PAPER_1929** — Inflation $N_{\text{efolds}} = A_5 = 60 +$ Theory of Permanence (companion integer-identity paper)
- **PAPER_1931** — $A_5 + SO_5 = 70$ cross-sector universality (companion integer-identity paper)
- **PAPER_1949** — F_{TRZ} Three-Face Framework (companion primitive-formalism paper)
- **PAPER_646** — Universal Inertial Operator (F_{UBii} coupling framework)
- **PAPER_1203** — Buoyancy primitive ($\beta_i = 0.6029$ canonical)
- **PAPER_1521** — D_{BSFG} derivative landmark
- **PAPER_1522** — K_{MEX} derivative landmark
- **PAPER_251 (Draft 2 candidate)** — Eta Carinae Homunculus DPM Invisibility (Great Eruption ~10-40 M_{sun} candidate for "40 M_{sun} stellar mass" row)
- **PAPER_424** — F_{UBii} Um Universal Companion Catalog (F_{UBii} domain reference, no explicit 40 identified)

9. Limitations + Open Questions

- Specific attributions for PAPER_1918's "40 M_{sun} " and "40 F_{UBii} coupling factor" catalog rows are TBD. A future PAPER_1970+ audit should identify the specific UQFF papers documenting these instances.
- The $r_c(\text{Virgo}) = 40$ kpc value in the source-code default is derived from Wolfram simulations of Virgo cluster physics; whether it survives higher-precision X-ray observations (Chandra, XRISM) with r_c stable at exactly 40 kpc is an observational question.
- Other cluster core radii (Perseus, Coma, Fornax) may or may not exhibit $r_c = 40$ kpc. PAPER_040 documents these clusters but does not explicitly extract 40 kpc from all four. Testable prediction: if all four exhibit $r_c \approx 40$ kpc, the identity is universal at cluster core scale; if only Virgo, it's Virgo-specific.
- The multi-scale correspondence (reactor Hz + cluster kpc + stellar mass + coupling factor) is currently a numerical observation. Whether it reflects a deeper "spatial $\times$ rotational" structural symmetry across ~30 orders of magnitude requires further first-principles derivation.

10. Revision Log

Draft 1 (2026-07-09): Initial write, positioned narrowly upfront given the honest-scholarship pattern established by PAPER_1965-1969 (all substantially walked back). Prior art acknowledged from the start:

- PAPER_1918 seminal for the multi-context catalog AND the "spatial $\times$ rotational" physical interpretation.
- PAPER_1472 seminal for the reactor 40 Hz specific instance.
- Round 107 stub upgrade attribution of the "40 kpc cluster core radius" catalog row to Virgo (via `VirgoExtXRayCalculator` source82_wolfram default).

The paper's contribution is limited to:

1. Specific anchor attribution for one PAPER_1918 catalog row (Virgo $\rightarrow$ 40 kpc).
2. Explicit cross-scale comparison table.
3. Anchor list for future catalog expansion.

Draft 1 does not claim novelty for the $D_{\text{phys}} \times SO_5 = 40$ identity itself, for its physical interpretation, or for either seminal instance (reactor Hz or the general multi-context pattern). If Drafts 2/3 find that specific-anchor attribution has already been done in another paper, further retreat may be needed.

Draft 2 (2026-07-09): Prior-art search for the two open PAPER_1918 catalog rows:

1. **"40 M_{sun} stellar mass" catalog row:** Candidate anchor identified — **PAPER_251 (Eta Carinae Homunculus DPM Invisibility)** states "The Great Eruption of ~1843 CE ejected ~10-40 M_{sun} in a bipolar" outflow. The upper bound of 40 M_{sun} matches the catalog entry. This is a candidate attribution

but not definitive (the 40 M_{sun} is an upper bound of an observed ejecta mass range, not an exact structural closure). If PAPER_1918's "40 M_{sun} stellar mass" refers to something more precise (e.g., specific massive-star canonical mass or IMF cutoff), that specific paper is not yet identified.

2. **"40 in F_UBii coupling factors" catalog row:** Specific paper not identified in Draft 2 search. PAPER_040 (F_UBii Virial Buoyancy at Perseus/Coma/Virgo) and PAPER_424 (F_UBii Um Universal Companion Catalog) are the F_UBii-specific papers but no explicit "40" appears as a coupling factor value. This row's specific attribution remains open.

Draft 3 (2026-07-09): Final honest scope. Cross-scale comparison table updated with Draft 2's Eta Carinae candidate for the 40 M_{sun} row and explicit "specific paper unidentified" note for the F_UBii row. Two of three PAPER_1918 catalog rows now have candidate attributions:

- 40 kpc cluster core radius → Virgo r_c (this paper, Round 107 confirmation via VirgoExtXRayCalculator)
- 40 M_{sun} stellar mass → Eta Carinae Great Eruption ejecta upper bound (Draft 2 candidate via PAPER_251)
- 40 F_UBii coupling factor → specific paper unidentified

The paper's contribution is thus a narrow attribution/documentation exercise: closing two of PAPER_1918's three catalog rows with candidate specific-paper anchors. The third row (F_UBii coupling factor 40) remains a research question for future work.

Reader takeaway: the $D_{\text{phys}} \times SO_5 = 40$ EXACT identity is a well-documented UQFF multi-context closure (PAPER_1918 seminal, PAPER_1472 reactor instance) whose specific per-instance anchors are now progressively being identified via double-check cycles. This paper contributes the Virgo r_c attribution (definitive) and the Eta Carinae M_{sun} attribution (candidate, requires verification). Future work should identify the F_UBii coupling factor 40 instance and audit whether Perseus/Coma/Fornax cluster cores also exhibit $r_c \approx 40$ kpc as a universal cluster property (per PAPER_040's four-anchor scope).

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